



GRAVITY-02

Class 09 - Science

1. How does the force of gravitation between two objects change when the distance between them is reduced to half? [2]
2. The earth and the moon are attracted to each other by gravitational force. Does the earth attract the moon with a force that is greater or smaller or the same as the force with which the moon attracts the earth? Why? [2]
3. Gravitational force on the surface of the moon is only $\frac{1}{6}$ as strong as gravitational force on the earth. What is the weight in newtons of a 10 kg object on the moon and on the earth? [2]
4. If gravitation is the power by which all bodies tend to fall towards each other, then why do all bodies tend towards the centre of the earth? [2]
5. A boy on a cliff 49 m high drops a stone. One second later, he throws a second stone after the first. They both hit the ground at the same time. With what speed did he throw the second stone? [2]
6. What is the magnitude of the gravitational force between the earth and a 1 kg object on its surface? (Mass of the earth is 6×10^{24} kg and radius of the earth is 6.4×10^6 m.) [2]
7. A stone is dropped from the edge of a roof. It passes a window 2 metre high in 0.1 second. How far is the roof above the top of the window? [2]
8. Why will a sheet of paper fall slower than one that is crumpled into a ball? [2]
9. What happens to the force between two objects, if [2]
 - i. The mass of one object is doubled?
 - ii. The distance between the objects is doubled and tripled?
 - iii. The masses of both objects are doubled?
10. What do you mean by acceleration due to gravity? [2]
11.
 - i. Name the SI units of thrust and pressure. [2]
 - ii. In which situation, do we exert more pressure on the ground when we stand on one foot or on both feet? Justify your answer.
12. Calculate the average density of the earth in terms of g , G and R . [2]
13. Give three differences between acceleration due to gravity (g) and universal gravitational constant (G). [2]
14. What is the importance of universal law of gravitation? [2]
15. You have a bag of cotton and an iron bar, each indicating a mass of 100 kg when measured on a weighing machine. In reality, one is heavier than other. Can you say which one is heavier and why? [2]
16. How does the force of attraction between the two bodies depend upon their masses and distance between them? [2]

A student thought that two bricks tied together would fall faster than a single one under the action of gravity. Do you agree with his hypothesis or not? Comment.
17. A firecracker is fired and it rises to a height of 1000 m. Find the [2]

- i. velocity by which it was released.
 - ii. time taken by it to reach the highest point. (Take, $g = 9.8 \text{ m/s}^2$)
18. A ball is thrown vertically upwards with a velocity of 49 m/s. Calculate [2]
- i. the maximum height to which it rises.
 - ii. the total time it takes to return to the surface of the earth.
19. A stone is thrown vertically upward with an initial velocity of 40 m/s. Taking $g = 10 \text{ m/s}^2$, find the maximum height reached by the stone. What is the net displacement and the total distance covered by the stone? [2]
20. A stone is allowed to fall from the top of a tower 100 m high and at the same time another stone is projected vertically upwards from the ground with a velocity of 25 m/s. Calculate when and where the two stones will meet. [2]
21. An object is thrown vertically upwards and rises to a height of 10 m. Calculate [2]
- i. the velocity with which the object was thrown upwards and
 - ii. the time taken by the object to reach the highest point.
22. Why is the weight of an object on the moon $1/6^{th}$ its weight on the earth? [2]
23. Suggest a method for calculating the mass of the moon. [2]
24. What is the source of the centripetal force that a planet requires revolving around the Sun? On what factors does that force depend? [2]
25. If the force of gravity somehow vanished away, why would we be sent flying in space? [2]
26. Two bodies of masses 3 kg and 12 kg are placed at a distance 12 m. A third body of mass 0.5 kg is to be placed at such a point that the force acting on this body is zero. Find the position of that point. [2]
27. Prove that if the earth attracts two bodies placed at the same distance from the centre of the earth with the same force, then their masses are equal. [2]
28. Write the formula to find the magnitude of the gravitational force between the earth and an object on the surface of the earth. [2]
29. Prove that if a body is thrown vertically upwards then the time of ascent is equal to the time of descent. [2]
30. Gravitational force acts on all objects in proportion to their masses. Why then, a heavy object does not fall faster than a light object? [2]
31. Why does a block of plastic released under water come up to the surface of water? [2]
32. Find the percentage change in the weight of a body, when it is taken from the equator to poles. The radius of the earth at poles is 6357 km and the radius at the equator is 6378 km. [2]
33. A stone is released from the top of a tower of height 19.6 m. Calculate its final velocity just before touching the ground. [2]
34. On a certain planet, a small stone tossed up at 15 m/s vertically upwards takes 7.5 s to return to the ground. What is the acceleration due to gravity on the planet? [2]
35. A ball is thrown vertically upwards with a velocity 'u'. Calculate the velocity with which it falls to the earth again. [2]
36. Two objects A and B begin a free fall from rest from the same height 1.00 s apart. How long after A was dropped will A and B be 10 m apart? [2]
37. A ball is thrown vertically upwards with a velocity of 49 ms^{-1} . Calculate: i) the maximum height to which it rises. ii) the total time it takes to return to the surface of the earth. [2]
38. State the universal law of gravitation. [2]
39. A solid weighs 100 gm in air and 88 gm when completely immersed in water. Calculate [2]

1. the upthrust
 2. the volume of the solid
 3. the relative density of the solid.
40. How does a force of gravitation between two objects change when the distance between them is reduced to half? [2]
 41. A bag of sugar weighs W at some place on the equator. If this bag is taken to Antarctica, will it weigh same, more or less? Give a reason for your answer. [2]
 42. Gravitational force on the surface of the moon is only $\frac{1}{6}$ as strong as gravitational force on the earth. What is the weight in newton of a 10 kg object on the moon and on the earth? [2]
 43. Why is the weight of an object on the moon $\frac{1}{6}$ th its weight on the earth? [2]
 44. A stone dropped from the roof of a building takes 4s to reach the ground. Calculate the height of the building. [2]
 45. An object is thrown vertically upwards and rises to a height of 13.07 m. Calculate [2]
 - i. the velocity with which the object was thrown upwards.
 - ii. the time taken by the object to reach the highest point.
 46. Why is it difficult to hold a school bag having a strap made of a thin and strong string? [2]
 47. When dropped from the same height a body reaches the ground quicker at poles than at the equator. Why? [2]
 48. Calculate the value of acceleration due to gravity g using the relation between g and G . [2]
 49. A ball is dropped from the top of a tower 40 m high. What is its velocity when it has covered a distance of 20 m? [2]

We would be its velocity when it hits the ground? (Take $g = 10 \text{ ms}^{-2}$)
 50. The acceleration due to gravity at the moon's surface is 1.67 ms^{-2} . If the radius of the moon is 1.74×10^6 , calculate the mass of the moon. Use the known value of G . [2]
 51. Mention any five phenomena that the universal law of gravitation was able to explain. [2]
 52. On the moon's surface, the acceleration due to gravity is 1.67 ms^{-2} . If the radius of the moon is $1.74 \times 10^6 \text{ m}$, calculate the mass of the moon. [2]
 53. How does the weight of an object vary with respect to the mass and radius of the earth? In a hypothetical case, if the diameter of the earth becomes half of its present value and its mass becomes four times of its present value, then how would the weight of any object on the surface of the earth be affected? [2]
 54.
 - i. Why does a bucket of water feel heavier when taken out of water? [2]
 - ii. Lead has greater density than iron and both are denser than water. Is the buoyant force on a lead object greater than, less than or equal to the buoyant force on an iron object of the same volume?
 55. In what sense does the moon fall towards the earth? Why does it not actually fall on the surface of Earth? [2]
 56. On the earth, a stone is thrown from a height in a direction parallel to the earth's surface while another stone is simultaneously dropped from the same height. Which stone would reach the ground first and why? [2]
 57.
 - i. Explain, why a completely immersed bottle in water bounces back on the surface? [2]
 - ii. Why does a bucket of water weigh less inside the well water?
 58. If the earth attracts an apple, does the apple also attract the earth? If so, why does the earth not move towards the apple? [2]
 59. Suppose that the radius of the earth becomes twice of its original radius without any change in its mass. Then what will happen to your weight? [2]
 60. A ball is thrown vertically upwards with a velocity of 49 m/s. Calculate [2]
 - (i) the maximum height to which it rises,
 - (ii) the total time it takes to return to the surface of the earth.

61. A ball is thrown up with a speed of 0.5 m/s. [2]
- i. How high will it go before it begins to fall?
 - ii. How long will it take to reach that height?
62. Write the formula to find a magnitude of the gravitational force between the earth and an object on the surface of the earth. [2]